



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

September 22, 2022

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 14856
(SEVENTH REVISION)

EXPIRATION DATE: 2026-08-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: BKC Industries, Inc.
Creedmoor, NC
2. PURPOSE AND LIMITATION:
 - a. This special permit authorizes the transportation in commerce of certain Division 2.1 and 2.2 materials in DOT 3A, 3AA, 3AX, 3AAX and 3T cylinders having water capacity over 125 lbs to be periodically requalified every ten years instead of every five years. The cylinders must be horizontally mounted on a motor vehicle (trailer chassis) or in an ISO frame or other framework of equivalent structural integrity. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in the development of this special permit only considered the hazards and risks associated with the transportation in commerce.
 - c. No party status will be granted to this special permit.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 180.209(a) and (b), in that cylinders with water capacity over 125 lbs may be requalified every ten years and they are not required to be removed from bundles or frames and hammer tested prior to each refill; and that the approved combination acoustic emission and ultrasonic examination (AE/UE) or 100% UE methods are authorized in lieu of hydrostatics testing and visual inspection.

NOTE: This does not relieve the holder of this special permit from securing an approval for retesting cylinders from the Associate Administrator for Hazardous Materials Safety.

Tracking Number: 2022054861

5. BASIS: This special permit is based on the application of BKC Industries, Inc. dated May 18, 2022, submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identi- fication Number	Packing Group
Air, compressed	2.2	UN1002	N/A
Argon, compressed	2.2	UN1006	N/A
Chlorodifluoromethane or Refrigerant Gas R22	2.1	UN1018	N/A
Compressed gas, flammable, n.o.s.	2.1	UN1954	N/A
Compressed gas, n.o.s.	2.2	UN1956	N/A
Cyclopropane	2.1	UN1027	N/A
1-Chloro-1.1-Difluoroethane or Refrigerant Gas R-142b	2.1	UN2517	N/A
1,1-Difluoroethylene or Refrigerant gas R 1132a	2.1	UN1959	N/A
Ethylene	2.1	UN1962	N/A
Helium, compressed	2.2	UN1046	N/A
Hexafluoroethane, or Refrigerant gas R-116	2.2	UN2193	N/A
Hydrogen, compressed	2.1	UN1049	N/A
Krypton, compressed	2.2	UN1056	N/A
Methane, compressed or Natural gas, compressed (<i>with high methane content</i>)	2.1	UN1971	N/A

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Neon, compressed	2.2	UN1065	N/A
Nitrogen, compressed	2.2	UN1066	N/A
Nitrogen trifluoride	2.2	UN2451	N/A
Nitrous oxide	2.2	UN1070	N/A
Oxygen, compressed	2.2	UN1072	N/A
Sulfur hexafluoride	2.2	UN1080	N/A
Tetrafluoromethane, or Refrigerant gas R 14	2.2	UN1982	N/A
Trifluoromethane or Refrigerant gas R 23	2.2	UN1984	N/A
Xenon	2.2	UN2036	N/A

7. **SAFETY CONTROL MEASURES:**

a. **PACKAGING:** Packaging prescribed are DOT 3A, 3AA, 3AX, 3AAX or 3T cylinders (tubes) mounted on a trailer chassis (tube trailers) or in an ISO frame, or other framework of equivalent structural integrity (see § 173.301(i)), that are subjected to periodic requalification and marking specified in §§ 180.205, 180.209, and 180.213 except that:

(1) Each cylinder mounted in a frame or on a trailer chassis, may be requalified every 10 years in accordance with § 180.205 using either the hydrostatic testing and visual inspection described in § 180.205 or in accordance with current DOT special permits applicable to these DOT specification cylinders using one of the following methods:

- 100% ultrasonic examination (UE) or
- Acoustic emission and ultrasonic examination (AE/UE)

The 10 year requalification period applies only to cylinders that meet all requirements specified in § 180.209(b) except that the cylinders may have a water capacity over 125 lbs and they are not hammer tested.

(2) Every five years, each cylinder including the cylinder neck and flange/sleeve must be subjected to an external visual inspection (without disassembly) in accordance with CGA Pamphlet C-6. A cylinder with excessive corrosion around the neck or under the flange/sleeve must be removed and examined in accordance with CGA Pamphlet C-23 ("Inspection of Tube Neck Mounting Surfaces."). BKC Industries, Inc. must record the results of the visual inspection for each cylinder or tube and submit the report to the Office of Hazardous Materials Safety (OHMS) upon request.

b. OPERATIONAL CONTROLS:

(1) Prior to use, the external neck threads and flange/sleeve of each tube with a diameter greater than or equal to 18" and mounted on tube trailers or modules must be inspected in accordance with CGA Pamphlet C-23. This inspection must be periodically repeated at least once every ten years.

(2) Before being used under the terms of this special permit, each DOT 3T cylinder (tube) must be subjected to shear wave ultrasonic examination (UE). The UE must be performed by an authorized DOT UE special permit holder or the cylinder (tube) manufacturer. Maximum acceptance defect (sidewall crack) for a tube that is tested by UE shear wave is 5% of the minimum design wall thickness of the tube. The UE shear wave must cover 100% of the sidewall of the cylinder.

(3) This special permit is limited to the DOT cylinders described in paragraph 7.a. above that are mounted horizontally in a suitable frame or on a trailer chassis. Flat bottom cylinders are not covered by this permit

(4) Maximum cycling (fillings) of each cylinder (tube) in a 10 year period must be less than or equal to 600 for DOT 3A, 3AA, 3AAX and 300 for DOT 3T cylinders. BKC Industries must record all fillings of each cylinder and submit the report to the OHMS upon request.

(5) All gases must have a dew point at or below minus 52 °F at one atmosphere.

(6) Hydrogen and hydrogen bearing gases are not authorized in DOT 3T specification cylinders (see § 173.302a(a)(4)).

(7) Carbon dioxide or any corrosive gas in any concentration shall not be added to these cylinders.

8. SPECIAL PROVISIONS:

a. A person who is not a holder of this special permit who receives a package covered by this special permit may reoffer it for transportation provided no modification or change is made to the package and it is reoffered for transportation in conformance with this special permit and the HMR.

b. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

c. MARKING:

(1) An exterior cylinder (tube) on each side of the frame or tube trailer must be plainly and durably marked "DOT-SP 14856" with the letters and numbers at least 2 inches in height on a contrasting background.

(2) The current retest date must be marked on the rear or side of the frame or tube trailer at approximately eye level. In the event the retest dates of the cylinders differ in the frame or tube trailer, the oldest retest date (meaning the date of the retest that must occur first) must be displayed.

d. Transportation of Division 2.1 (flammable gases) is not authorized aboard a cargo vessel unless specifically authorized in the Hazardous Materials Table (§ 172.101).

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, cargo vessel and rail freight.10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel or motor vehicle used to transport packages covered by this special permit.11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)—“The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

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